$Z_C = \frac{1}{j\omega C}$

$Z_I = j\omega L$

Dylan Thornburg

9.30 $v_s(t) = 25\sin(4000t)\ V$

$v_s(t) = 25\cos(4000t - 90)\ V$

$4000(.0025)j = 10j\ \Omega$

$\frac{1}{j(4000)(.00000125)}\ mH = -20j\ \Omega$

$V_s(t) = 25\angle -90$

$5 + (10j \,||\, 10 + (20 \,||\, -20j))$

$(10 + -40i)$

$Z = 10 + 10i$

$\frac{25\cos(4000t - 90)}{10 + 10i}$

$\frac{25\angle -90}{10 + 10j} = 1.77\angle -135^0$

$i_o(t) = 1.77$

$\boxed{i_o(t) = 1.77\cos(4000t - 135)\ A}$

9.32

$V_b = 50 - j25\ V \quad I_b = 25\angle 0$

$I_a = 50 - j25 / 500 + 250j = 60 - j80\ mA = 100\angle -53.13\ mA$

$I_c = 50 - j25 + j50 / 1000 = 50 + j25\ mA = 55.9\angle 26.57\ mA$

$I_g = 100\angle -53.13 + 55.9\angle 26.57 + 25\angle 0 = 145.77\angle -22.17\ mA$

$i_g = 100\cos(1500t - 53.13^\circ)$ mA

$i_C = 55.9\cos(1500t + 26.57^\circ)$ mA

$i_g = 145.7\cos(1500t - 22.17^\circ)$ mA

9.35 $V_g = 25\angle 0$ V $\quad I_g = 5\angle 90$ A

$V_1 = 10$ V $\qquad \frac{15}{4-j3} = 2.4 + j1.8$

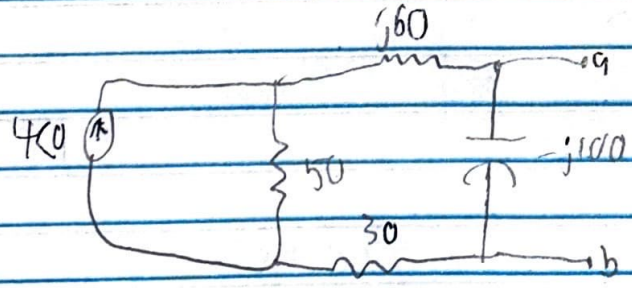
$I_b = (2.4 + j1.8) - j5 = \boxed{2.4 - j3.2 A}$

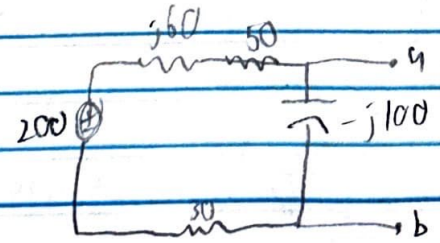
$V_2 = -j5(2.4 - j3.2) + (4-j3)(2.4 + j1.8) = -1 - j12$ V

$I_2 = (6.2 - j6.6) - (2.4 - j3.2) = 3.8 - j3.4$

$Z = \frac{V_2}{I_2} = \frac{-1-j12}{3.8-j3.4} = \boxed{1.42 - j1.88\,\Omega}$

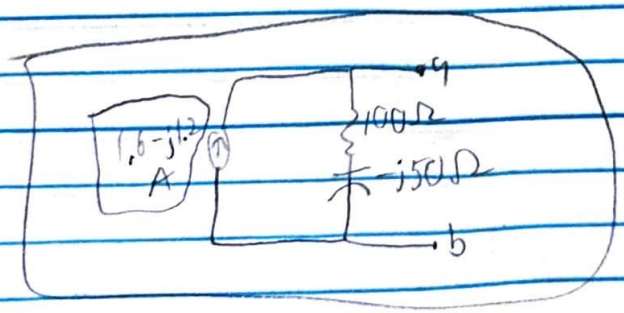
9.44





$\frac{200}{j60+80} = 1.6 - j1.2$

$-j100 \| (80 + 60j) = 100 - 50j$



9.45

$50\cos(400t-90)V$

$50\angle -90$

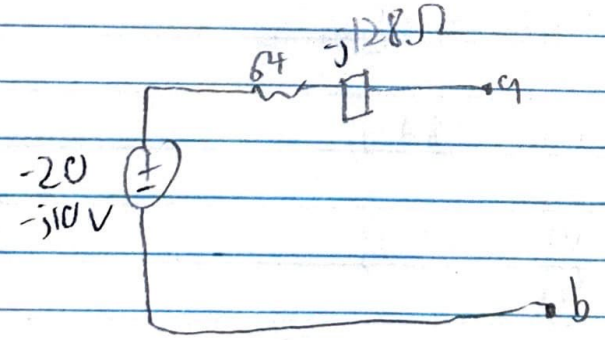
$$\frac{1}{j400(.00003125)} = -j80\,\Omega$$

$$.4(400)j = 160j\,\Omega$$

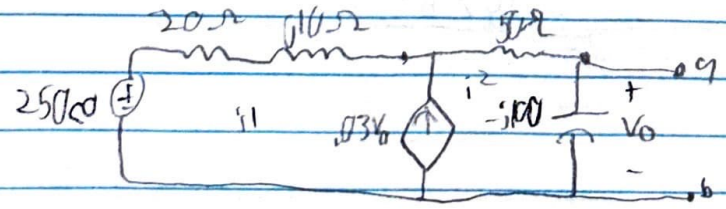
$$320 \parallel -j80 \parallel j160 = 64 - j128\,\Omega$$

$$V_{a-b} = \frac{-j80 \parallel j160}{320 + j160 \parallel -j80}(-j50) \qquad \frac{1}{-j80} + \frac{1}{j160}$$

$$\boxed{V_{a+b} = -20 - 10j}$$



9.50



$$\frac{250}{20+j10} = 3.5 - j.5A$$

$$\frac{V_1 - 250}{20+j10} - .03V_0 + \frac{V_1}{50-j100} = 0$$

$$\frac{300-j400}{3.5-j.5} = \frac{100-j100}{}$$

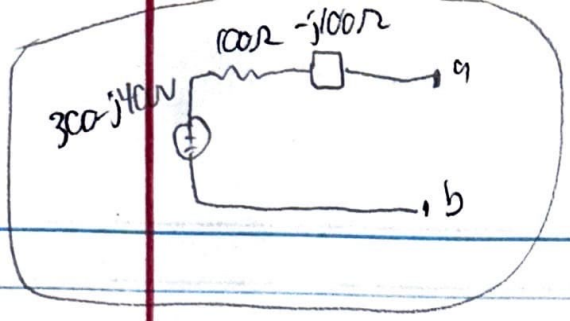
$$V_0 = \frac{-j100}{50-j100}V_1$$

Ans on back →

$$\frac{V1}{20+j10} + \frac{.03V_0}{50-j100} + \frac{V_1}{50-j100} = \frac{250}{20+j10}$$

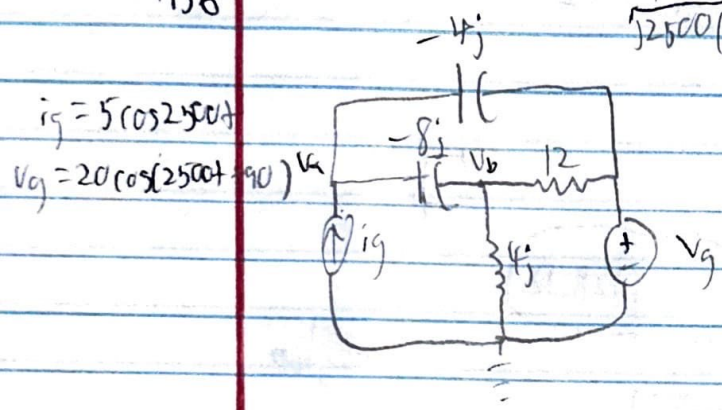
$V_0 = 300 - j400V$

$V_1 = 500 - j250V$



9.56

$$\frac{1}{j2500(.0001)} = -4j \cdot 2 = -8j$$



$i_g = 5\cos 2500t$

$v_g = 20\cos(2500t + 90)$ V

$$j(2500)(.0016) = 4j$$

$$-i_g + \frac{V_a - V_b}{-8j} + \frac{V_a - V_g}{-4j} = 0$$

$$-5 + \frac{V_a - V_b}{-8j} + \frac{V_a - 20j}{-4j} = 0$$

$$\frac{V_b - V_a}{-8j} + \frac{V_b}{4j} + \frac{V_b - 20j}{12} = 0$$

$$V_a = \frac{-8/3 + 4j}{3}$$

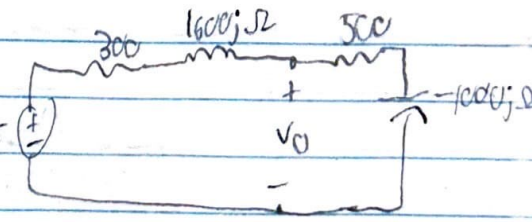
$$V_b = -8 + 4j$$

$$\frac{V_b}{4j} = -1 - 2j$$

$$2.24\angle 63.43^\circ$$

$$\boxed{i_o = 2.24\cos(2500t + 63.43)}$$

9.65



$V_g = 100\cos 8000t$

$$(100)\frac{500 - 1000j}{300 + 1600j + 500 - 1000j} = -20 - 110j$$

$$\downarrow$$

$$111.8\angle -100.3^\circ$$

$$\boxed{V_o(t) = 111.8\cos(8000t - 100.3)\text{ V}}$$